databricks

DSL Lite

Databricks Security Lakehouse

Cybersecurity Data & AI Engineering Accelerator · Medallion Architecture · OCSF 1.7.0

Databricks Forward Deployed Engineering
Built for the Cyber Lakehouse

DSL Lite is a declarative, configuration-driven framework that transforms raw security logs from any source into **OCSF-compliant, analytics-ready tables** on Databricks — with no custom Spark code required. Security teams get a production-grade Bronze → Silver → Gold medallion pipeline in hours, not weeks or months. *A reusable, scalable blueprint to build a Cyber Lakehouse.*

MEDALLION ARCHITECTURE

Bronze — Ingest & Amend
Auto Loader streaming ingestion. Raw payload preserved as Variant or text with *source*, *sourcetype*, event *time*, and auto-injected file metadata.

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Silver — Parse & Curate
Typed fields flattened from Variant or extracted as regex patterns into a structured schema. Filters, temporary columns, and business logic optionally applied.

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Gold — Map & Normalize
OCSF-compliant Delta Sinks with enforced schema standardization. Cross-vendor analytics on unified event tables with full source lineage tracking.

KEY FEATURES

- **Zero Spark code** — declarative YAML presets define the full end-to-end pipeline
- **OCSF 1.7.0 compliance** — unified Common Information Model across all vendors for cross-source security analytics
- **OCSF categories & events** — Network, System, IAM, Findings, Application Activity · classes: DNS, HTTP, File System, Authentication, Process, API, and more
- **Source lineage tracking** — *source*, *sourcetype*, and *time* carried through every layer
- **Multi-format ingestion** — JSON, CSV, Syslog, and text via Auto Loader as Variant data type or text format
- **Streaming & batch** — continuous or trigger-based pipeline execution
- **Schema enforcement** — typed, versioned OCSF gold tables for performance and governance
- **Lookup enrichment** — enrich events with Delta or CSV reference tables
- **Liquid Clustering** — time-based clustering optimized for security queries
- **Idempotent ingestion** — unique *record_id* per event prevents duplicate records across pipeline reruns
- **One-command deployment** — Declarative Automation Bundles (DABs) per source
- **Agentic development** — built-in agent skills for Databricks Genie and AI coding tools

EXECUTION MODES

Spark Declarative Pipelines SDP
Lakeflow-managed pipeline. Serverless, continuous or triggered. Gold tables as Delta Sinks for cross-vendor consolidation.

Spark Structured Streaming SSS
Classic compute job. Three independent tasks (bronze / silver / gold) or a single medallion job. Full control over cluster configuration.

Streaming ingest (on request): Native file-based ingestion via Auto Loader is the default. Streaming source support — **Apache Kafka, Databricks Zerobus Ingest, Azure Event Hubs, AWS Kinesis** — are available as a development feature and can be enabled per engagement. Contact your Databricks team to scope.

SAMPLE PRESETS & OCSF GOLD OUTPUTS

Cisco IOS
authentication · authorize_session · network_activity · process_activity

Cloudflare Gateway DNS
dns_activity

GitHub Audit Logs
authentication · account_change · authorize_session · user_access · group_management · entity_management · api_activity

Okta System Log
authentication · account_change · group_management

Zeek Conn
network_activity

AWS VPC Flow Logs
network_activity

New sources are added by authoring a single *preset.yaml* — no framework changes required. Ingest from landing zone — S3, ADLS Gen2, GCS, Databricks Volumes, or DBFS.

Additional vendors available for preset development: Palo Alto Networks · Microsoft Azure / Entra ID · Zscaler (ZIA / ZPA) · CrowdStrike Falcon · Atlassian · Slack · SentinelOne · Fortinet · Check Point · AWS CloudTrail · Google Cloud · Microsoft 365 · Microsoft Defender · Carbon Black · Splunk (as source) · and more

DEVELOPER EXPERIENCE

🤖 **Preset Agent**
Generates a first-draft *preset.yaml* from a Unity Catalog table or raw logs using Databricks Foundation Models — entirely inside the workspace.

🔍 **Preset Explorer**
Iterate on presets against real data in batch mode. Preview all three layers before deployment.

📊 **Pipeline Profiler**
Schema diff, null-rate comparison, end-to-end sample run, and OCSF coverage checks. Saves a Markdown report to a Volume.

Agentic development: Built-in agent skills augment AI coding tools (Claude Code, Cursor, Copilot) and **Databricks Genie** for natural-language preset authoring, pipeline Q&A, and OCSF mapping assistance.

PLATFORM REQUIREMENTS

Databricks Runtime · Unity Catalog · Serverless (recommended) · Databricks Genie (recommended) · SDP — Spark Declarative Pipelines · SSS — Spark Structured Streaming · Git Integration · Databricks CLI · Databricks Auto Loader · Apache Spark · PyYAML · Delta Lake

USE CASES

SOC Operations · Threat Hunting · Compliance Reporting · UEBA & Behavior Analytics · Security Data Engineering · Anomaly Detection · Security Monitoring · Dashboards & Diagnostics · Digital Forensics & Incident Response · Insider Threat Detection · IOC Matching · M-21-31 Compliance